Multimodal Encoder and Input-Pipeline Optimization

Deep Dive into Kimi K3 — Chapter 40

Yin Yang

Foundation Books Project

The one idea of this chapter

A multimodal step can stall before the backbone sees a token: readiness, not arithmetic, decides when useful work resumes.

The one idea of this chapter

A multimodal step can stall before the backbone sees a token:
readiness, not arithmetic, decides when useful work resumes.

Goal: count each pressure separately, pack and shard without changing the computation, place encoder work only where it is ready and fits, and judge every intervention on the complete step.

Roadmap

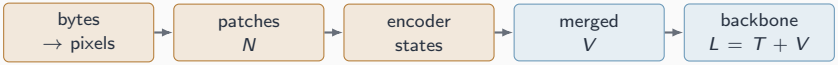
Inputs, identity, and workload

Parallel encoder work and bubbles

Services, memory, and the whole step

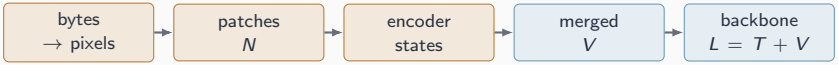
Inputs, identity, and workload

From media bytes to typed pressures



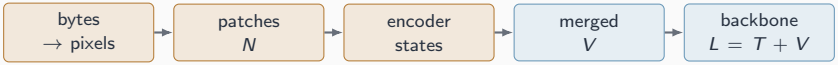
- A–D fixture: text, 56^2 image, 112^2 image, four-frame video; L totals 196. [DERIVED: synthetic fixture]

From media bytes to typed pressures



- A–D fixture: text, 56^2 image, 112^2 image, four-frame video; L totals 196. [DERIVED: synthetic fixture]
- C and D both have 64 encoder positions, but 4096 versus 1024 spatial pairs — a length-only balancer can worsen the visual bottleneck.

From media bytes to typed pressures



- A–D fixture: text, 56^2 image, 112^2 image, four-frame video; L totals 196. [DERIVED: synthetic fixture]
- C and D both have 64 encoder positions, but 4096 versus 1024 spatial pairs — a length-only balancer can worsen the visual bottleneck.
- Packing is a bijection carrying identity, masks, and targets; index 36 (C) must not see index 35 (B).

Parallel encoder work and bubbles

Shard queries, gather K/V, place only ready work

- Vision CP: each query shard needs every key and value of its domain; backward sums K/V contributions and returns owner slices. [REPORTED: K3 Sec. 5.2.3]

Shard queries, gather K/V, place only ready work

- Vision CP: each query shard needs every key and value of its domain; backward sums K/V contributions and returns owner slices. [REPORTED: K3 Sec. 5.2.3]
- Group size: $12/p + g(p)$ gives 12, 8, 8 — four ranks buy nothing over two. [DERIVED: declared model]

Shard queries, gather K/V, place only ready work

- Vision CP: each query shard needs every key and value of its domain; backward sums K/V contributions and returns owner slices. [REPORTED: K3 Sec. 5.2.3]
- Group size: $12/p + g(p)$ gives 12, 8, 8 — four ranks buy nothing over two. [DERIVED: declared model]
- Encoder backward needs its own sample's $\overline{z}$; replay must reuse the same closure, adding no second loss.

Shard queries, gather K/V, place only ready work

- Vision CP: each query shard needs every key and value of its domain; backward sums K/V contributions and returns owner slices. [REPORTED: K3 Sec. 5.2.3]
- Group size: $12/p + g(p)$ gives 12, 8, 8 — four ranks buy nothing over two. [DERIVED: declared model]
- Encoder backward needs its own sample's $\overline{z}$; replay must reuse the same closure, adding no second loss.
- A bubble is usable only if a ready job fits a common window across all its resources before its deadline.

Services, memory, and the whole step

Bounded overlap, judged on the complete step

- Byte-credit admission: a 12 MiB queue defers a 9 MiB object until the 6 MiB one leaves; deferred objects still count upstream.

Bounded overlap, judged on the complete step

- Byte-credit admission: a 12 MiB queue defers a 9 MiB object until the 6 MiB one leaves; deferred objects still count upstream.
- Early production can fail memory: $6 + 9 + 3 = 18 > 12$ MiB; the delayed placement fits and delivers at 16.

Bounded overlap, judged on the complete step

- Byte-credit admission: a 12 MiB queue defers a 9 MiB object until the 6 MiB one leaves; deferred objects still count upstream.
- Early production can fail memory: $6 + 9 + 3 = 18 > 12$ MiB; the delayed placement fits and delivers at 16.
- Complete step: exposed 34, one-rank overlap 33, two-rank C 34, late C 44. [DERIVED: finite model, not hardware]

Bounded overlap, judged on the complete step

- Byte-credit admission: a 12 MiB queue defers a 9 MiB object until the 6 MiB one leaves; deferred objects still count upstream.
- Early production can fail memory: $6 + 9 + 3 = 18 > 12$ MiB; the delayed placement fits and delivers at 16.
- Complete step: exposed 34, one-rank overlap 33, two-rank C 34, late C 44. [DERIVED: finite model, not hardware]
- Equivalence needs one common weighted denominator, plus oracles and deliberate failures.

Bounded overlap, judged on the complete step

- Byte-credit admission: a 12 MiB queue defers a 9 MiB object until the 6 MiB one leaves; deferred objects still count upstream.
- Early production can fail memory: $6 + 9 + 3 = 18 > 12$ MiB; the delayed placement fits and delivers at 16.
- Complete step: exposed 34, one-rank overlap 33, two-rank C 34, late C 44. [DERIVED: finite model, not hardware]
- Equivalence needs one common weighted denominator, plus oracles and deliberate failures.

Validate identity before speed. Next: Chapter 41, Checkpoint Engine and Online Weight Publication.